

Responsibility and verification of AI-generated code in online discussions

Group 36 | COMP3020 Social Web Analytics | Hacker News discussion corpus

Overall question

How are responsibility and verification of AI-generated code discussed online, and how are these discussions structured through reply interactions?

5
HN discussion
threads

313
raw comments

278
clean comments

186
authors

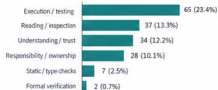
DATA + METHODS

- Source: Hacker News API items from five selected threads.
- Cleaned dead/deleted comments: 313 raw → 278 retained.
- Reply outcome: a retained comment has at least one retained direct child.
- Edges use real parent_id/comment_id reply links only.
- Analyses: word frequency, chi-square test, clustering, directed reply network.

collect → clean → code → analyse

RQ1 WHAT IS DISCUSSED?

Dictionary-assisted description



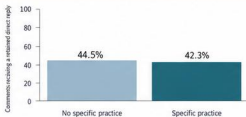
Comments with a candidate mention (non-exclusive)

Most visible candidate indicators:

- Execution / testing: 65 comments (23.4%).
- Reading / inspection: 37 comments (13.3%).
- Understanding / trust: 34 comments (12.2%).
- Responsibility / ownership: 28 comments (10.1%).
- Dictionary matches are screening summaries, not proof of endorsement.

Takeaway: the corpus discusses testing, review, ownership and trust, but context matters.

RQ2 HYPOTHESIS TEST

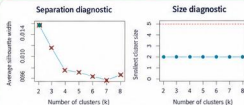


Specific verification mentions were not associated with receiving a direct reply.

2x2 table: No practice 86 no reply / 69 reply; Specific practice 71 no reply / 52 reply.

$\chi^2(1) = 0.064, p = 0.801 \rightarrow$ no statistically significant association

RQ3 CONTENT CLUSTERS



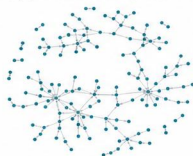
TF-IDF + cosine distance + average-link hierarchical clustering

- Selected $k = 2$ only as a diagnostic fallback.
- Mean silhouette = 0.015: very weak separation.
- Smallest cluster size = 2 across tested k .
- Conclusion: vocabulary did not form clear, balanced thematic groups.

This weak result is still useful: it warns us not to over-claim clusters as stable themes.

RQ4 DIRECT-REPLY NETWORK

Full reply network: 144 authors, 193 directed pairs



Arrow: replying author → parent author; nodes have equal size.

144 authors in graph, 193 directed author pairs, 128 largest weak component

Edge A → B means author A directly replied to a comment by author B.

Top in-degree: **simonw = 14**, **fish toaster = 14**. Centrality indicates structural position, not expertise.



LIMITATIONS

What we can say

- The selected Hacker News corpus shows how participants discuss responsibility and verification.
- Network edges are real reply relations from the data.

What we cannot say

- Not representative of all programmers or all Hacker News discussions.
- Cannot infer causation or real-world code quality.
- Manual TRUE/FALSE coding involved judgement; comments in the same thread are not fully independent.

Bottom line: public discussion links AI-code verification with responsibility, but the strongest evidence here is descriptive and bounded.

MAIN FINDINGS

- Verification is discussed through testing, review and responsibility.
- Specific verification mentions did not receive more replies in this sample.
- Clustering produced weak separation, so theme claims must stay cautious.
- The reply network is sparse but mostly connected through one large weak component.
- Receiving replies is not the same as agreement, quality or influence.

Story of the poster

The project connects what people say about checking AI-generated code with how others reply to those comments.

GitHub repo

github.com/ledinhtrung05-coder/COMP3020-Group36